# Arturo Sauza de la Vega, Ph.D.

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## Employment History

2025 – · · · · **Postdoctoral Scholar,** University of Iowa.
2024 **Graduate Research Fellow.** Theoretical Division, T-1, Los Alamos National Laboratory.

2020 **Lecturer.** School of Chemistry, National Autonomous University of Mexico.
2019 **Visiting Research Fellow.** The University of Manchester.

## Education

2020 – 2025 **Ph.D. in Chemistry, University of Chicago.**
Thesis title: *Multireference Studies of Transition Metal and Actinide Containing Molecular Systems.*

2020– 2021 **M.Sc. in Chemistry, University of Chicago.**

2018 – 2020 **M.Sc. in Chemistry, National Autonomous University of Mexico (UNAM, Mexico).**
Thesis title: *Non-additive effects in the interaction of water clusters with mono- and divalent cations and anions.*

2010 – 2017 **B.Sc. in Chemistry, National Autonomous University of Mexico (UNAM, Mexico).**
Thesis title: *Water clusters as bifunctional catalysts in organic chemistry. Ethylene oxide hydrolysis.*

2009 – 2015 **B.Eng. in Industrial Chemical Engineering, National Polytechnic Institute (IPN,Mexico).**
Thesis title: *Electron density topology and analysis of noncovalent interactions of dehydroalanines relevant in medicinal chemistry.*

## Research Publications

1. Vallejo Narváez, W. E.; Jiménez, E. I.; Romero-Montalvo, E.; **Sauza-de la Vega, A.**; Quiroz-García, B.; Hernández-Rodríguez, M.; Rocha-Rinza, T. Acidity and basicity interplay in amide and imide self-association. *Chem. Sci.*, **2018**, 9, 4402–4413. DOI: 10.1039/C8SC01020J.

2. **Sauza-de la Vega, A.**; Guevara-Vela, J. M.; Rocha-Rinza, T. Cooperativity and anticooperativity in ion-water interactions. Implications in the aqueous solvation of ions. *ChemPhysChem*, **2021**, 22, 1269–1285. DOI: 10.1002/cphc.202000981.

3. **Sauza-de la Vega, A.**; Salazar-Lozas, H.; Vallejo Narváez, W. E.; Hernández-Rodríguez, M.; Rocha-Rinza, T. Water clusters as bifunctional catalysts in organic chemistry. The hydrolysis of oxirane and its methyl derivatives. *Org. Biomol. Chem.*, **2021**, 19, 6776–6780. DOI: 10.1039/D1OB01026C.

4. **Sauza-de la Vega, A.**; Duarte, L. J. ; Silva, A. ; Skelton, J. ; Rocha-Rinza, T. ; Popelier, P. L. A., Towards

and Atomistic Understanding of Polymorphism in Molecular Solids. *Phys. Chem. Chem. Phys.*, **2022**, 24, 11278–11294. DOI: 10.1039/d2cp00457g.

5. Aristizabal-Ferreira, V. A.; Guevara-Vela, J. M.; **Sauza-de la Vega, A.**; Martín Pendás, Á.; Fuentes-Pineda, G., Rocha-Rinza, T., Computation of photovoltaic and stability properties of hybrid organic-inorganic perovskites via convolutional neural networks. *Theor. Chem. Acc.*, **2022**, 141, 19. DOI: 10.1007/s00214-022-02875-9.

6. **Sauza-de la Vega, A.**; Pandharkar, R. ; Stroscio, G. D. ; Sarkar, A.; Truhlar, D. G.; Gagliardi, L., Multiconfiguration Pair-Density Functional Theory for Chromium(IV) Molecular Qubits. *JACS Au*, **2022**, 2, 2029–2037. DOI: 10.1021/jacsau.2c00306.

7. Guevara-Vela, J. M.; **Sauza-de la Vega, A.**; Gallegos, M. Martín Pendás, Á. ; Rocha-Rinza, T., Wave function analyses of scandium-doped aluminum clusters, $Al_n$Sc ($n = 1 - 24$), and their $CO_2$ fixiation abilities. *Phys. Chem. Chem. Phys.*, **2023**, 25, 18854–18865. DOI: 10.1039/D3CP01730C.

8. Kunstelj, Š.; Darù, A.; **Sauza-de la Vega, A.**; Stroscio, G. D.; Edwards, E.; Gagliardi, L.; Wuttig, A., Competitive Valerate Binding Enables $RuO_2$-Mediated Butene Electrosynthesis in Water. *J. Am. Chem. Soc.*, **2024**, 146, 20584–20593. DOI: 10.1021/jacs.4c01776.

9. Hertler, P..; **Sauza-de la Vega, A.**; Darù, A.; Sarkar, A.; Lewis, R.A.; Gagliardi, L.; Hayton, T.W., A Homoleptic Fe(IV) Ketimide Complex with a Low-Lying Excited State. *Chem. Sci.*, **2024**, 15, 16559-16566. DOI: 10.1039/D4SC04880F.

10. Kirlikovali, K. O.; Gómez-Torres, A.; **Sauza-de la Vega, A.**; Darù, A.; Krzyaniak, M. D.; Wasielewski, M. R.; Gagliardi, L.; Farha, O. K., Electronically Tunable Low-Valent Uranium Metallacarboranes. *Inorg. Chem.*, **2025**, 64, 4749-4760. DOI: 10.1021/acs.inorgchem.4c04431

11. **Sauza-de la Vega, A.**; Darù, A.; Nofz, S. Gagliardi, L.; Designing Molecular Qubits: Computational Insights into First-Row and Group 6 Transition Metal Complexes. *Chem. Sci.*, **2025**, 16, 12896-12905, DOI: 10.1039/D5SC02544C.

12. **Sauza-de la Vega, A.**; Beltrán-Leiva, M. J.; Gagliardi, L.; Batista, E.R.; Yang, P.; Chemical Bonding and Oxidation State Variability in Lanthanide and Actinide Systems with Non-Innocent Ligands. **2026**, *To be submitted for publication* .

13. Gullett, K. L.; Silva, C. L.; **Sauza-de la Vega, A.**; Shaw, T. E.; Pereiro, F. A.; Collins, T. S.; Coughlin, E.J.; Lapsheva, E.; Anderson, N. H.; Gaggioli, C.; Zeller, M.; Mullis, M. S.; Shafer, J. C.; Kozimor, S. A.; Gagliardi, L.; Schelter, E. J. .; Bart, S.; Isolation and Characterization of a Pentakis(imido)uranate (VI) Tetraanion. **2026**, *Under Review* .

## Conferences and Presentations

- **XVI International Symposium "Universities Contributions to Teaching, Research and Technology Development", September 2015, Mexico City, Mexico.** Oral Presentation: Dehydroalanine derivatives conformational analysis using computational quantum chemistry.

- **Symposium "Frontiers in Computational Chemistry 2016", August 2016, Mexico City, Mexico.** Poster: Topology of the electron density and non-covalent interactions analysis of dehydroalanine derivatives relevant in medicinal chemistry.

- **XV Mexican Meeting of Theoretical Physical Chemistry, November 2016, Mérida, México** Poster: Topology of the electron density and non-covalent interactions analysis of dehydroalanine derivatives relevant in medicinal chemistry.

- **XVI Mexican Meeting of Theoretical Physical Chemistry, November 2017, Puebla, Mexico** Poster: Water clusters as bifunctional catalysts in Organic Chemistry. Ethylene Oxide hydrolysis.

- **XVII Mexican Meeting of Theoretical Physical Chemistry, November 2018, Monterrey, Mexico** Poster: Study of the hydrogen bond in water clusters with cations and anions through the IQA energy partition.
- **American Chemical Society 2021 Fall Meeting** Poster: Metal aryl complex candidates for molecular qubits.
- **American Chemical Society Spring Meeting, March 2022, San Diego, CA, United States of America** Oral Presentation: Zero-Field Splitting of Cr(IV) Molecular Qubits: Theoretical Insights.
- **Physical & Environmental Seminar, September 2025, Iowa City, IA, United States of America** Oral Presentation: Multiconfiguration Pair-Density Functional Theory for Molecular Qubits.
- **3rd International Workshop on Theory Frontiers in Actinide Science: Chemistry & Materials, March 2026, Seattle, WA, United States of America** Oral Presentation: Steric-Induced Variations in f-Element Reactivity with Bulky Terphenylborohydride Ligands.

## Skills

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| Languages | Strong reading, writing and speaking competencies for English, and Spanish (Native). |
| Coding | Bash, Python. |
| Scientific Software | : Molcas, Gamess-US, Gaussian 16, Aimall, Promolden, Gnuplot, Vesta, VMD, Orca, ADF, PySCF, Turbomole. |
| Additional software | : LaTeX, Tikz, Gimp, Inkscape, Microsoft Office. |

## Miscellaneous Experience

### Awards and Achievements

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| 2018 | **CONACyT, Mexico**, Scholarship for graduate studies at the National Autonomous University of Mexico Chemistry Graduate program. |
| 2019 | **CONACyT, Mexico**, Summer research fellowship at The University of Manchester. |
| 2020 | **University of Minnesota**, SURF-CTC Summer Research Fellowship. |
| | **University of Minnesota**, Departament of Chemistry Fellowship (Not Accepted). |
| 2024 | **Los Alamos National Laboratory**, Glenn T. Seaborg Fellowship for Graduate Research Internship. |
| 2026 | **American Chemical Society**, ACS Graduate Student and Postdoctoral Scholars Recognition Program. |

## References


**Prof. Tomás Rocha Rinza**
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